

PROJECT ABSTRACT

AI-Powered Vision Robot for Product Handling

Dark Stores

“AI-Powered Vision Robot for Product Handling - Dark Stores, presents an autonomous robotic system designed to automate the identification, retrieval, movement, and collection of products within a dark-store environment. Dark stores are fulfillment-oriented facilities where products are arranged at predefined locations and orders are prepared without conventional customer movement inside the store. The proposed system integrates embedded systems, computer vision, machine learning, wireless communication, web development, and robotic handling into a unified product-handling platform. The robot operates on a compact four-wheel chassis with ESP32-based control, IR sensors for line following, a camera module for product recognition, a servo-driven robotic arm, and a precision gripper. The system receives product information from a web application and uses this information to determine the sequence in which the required products need to be collected.

The proposed system addresses the need for reducing manual effort and improving the efficiency of repetitive product retrieval operations in dark-store environments. The robot is intended to identify the required products at predefined storage locations, verify them using computer vision, and transfer them to a basket attached to the robot. A web application is used to provide product selection and order management, while APIs are used for communication between the digital order system and the robotic platform. The system is designed to support controlled and repeatable product retrieval with minimum human intervention, while also providing information about unavailable products and the final order status.

The major objectives of the project are to develop a web-based order interface for product selection and order generation; establish wireless communication between the web application and the robotic system; implement line-following navigation using IR sensors and ESP32-based control; detect and recognize selected products using a camera module and machine-learning-based computer vision; perform product-specific robotic arm and gripper operations for reliable handling; optimize the retrieval sequence based on predefined product locations; identify unavailable products through visual verification; and generate an updated order summary and bill based on the products successfully retrieved.

The robot operates on a predefined line-following path, making it suitable for dark-store environments where storage positions remain fixed or semi-fixed. IR sensors positioned beneath the chassis continuously provide path information to the motion controller, while the ESP32 controls the wheel motors through a suitable motor driver. Each product location is predefined using a known position or distance from the starting point. When the robot reaches a designated product station, it stops for a specified period to allow product detection and handling. The camera module captures the product, and the trained recognition model determines whether the expected product is present. The system is trained using some selected products representing variations in shape, color, size, and surface appearance. A suitable machine-learning platform can be used to develop the recognition model, which is then integrated into the robotic vision subsystem.

After successful product identification, the robotic arm moves to the predefined pickup position and the gripper performs the required handling operation according to the recognized product. Product-specific servo positions and gripping parameters can be configured to accommodate differences in product dimensions and shape. After pickup, the arm transfers the product to the basket mounted on the robot chassis and returns to its default position. The robot then continues toward the next required product. This sequence of navigation, detection, verification, handling, transfer, and placement enables repeated product retrieval in a controlled manner.

An additional feature of the system is retrieval-sequence optimization. Since users may select products in any order, the system can determine an efficient sequence using the known positions and path distances of the products. A suitable route-optimization approach can be used to reduce unnecessary movement and improve the overall retrieval process. The system also includes a mechanism for handling unavailable or missing products. When the requested product is not detected at its designated location within the specified verification period, the robot classifies it as unavailable and displays an appropriate status message on an OLED display. The order information can also be updated through the web application so that unavailable products are clearly separated from the successfully retrieved items.

The expected outcome of the project is a working prototype capable of receiving product orders through a web application, communicating the order to the robotic system through wireless connectivity and APIs, navigating along predefined paths, recognizing selected products using computer vision and machine learning, performing automated product handling, identifying unavailable items, and generating an updated order summary and bill. The proposed system provides a structured digital-to-physical workflow for dark-store product fulfillment and demonstrates the practical integration of web development, IoT communication, embedded control, computer vision, machine learning, and robotic handling into an automated product-handling framework.

UNDER THE GUIDANCE OF

Dr. T.A.N. Surya Narayana Varma

M.Tech., Ph.D.

Professor, Dept of ECE

MVGR College of Engineering(A)

Vizianagaram, Andhra Pradesh

SUBMITTED BY

Batch 3, ECE-B

K. Jyothi -23331A0483

Md. Azmath Ali - 23331A04B0

N. Naveen - 23331A04B8

K. Ganapathi Naidu - 23331A0486

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